

TRANSPORT ACROSS CELL MEMBRANE (BULK TRANSPORT)

Dr. Fatima Haq
Lecturer
DIKIOHS, DUHS.

TRANSPORT MECHANISMS

- Active process

- Primary Transport

- Secondary
Transport

- Passive process

- Simple diffusion

- Facilitated diffusion

- Osmosis

- Bulk flow

- Filtration

PASSIVE TRANSPORT

- Movement of molecules through the membrane in which no energy is required from the cell.
- Molecules move in response to a concentration gradient.
- *A concentration gradient is a difference between the concentration on one side of the membrane and that on the other side.*
- Passive transport mechanisms only move substances along the concentration gradient.

VESICULAR TRANSPORT

- It is the transport of membrane bounded substances moving across plasma membrane
- It is classified into:
 - 1. Endocytosis
 - 2. Exocytosis.

BULK TRANSPORT

Bulk transport is a type of transport which involves the transport of large amount of substance like lipid droplets and solid food particles across plasma membrane by utilizing energy.

Eg :

Substances that can move via bulk transport are like hormones, polysaccharides, etc. An example of this is the engulfing of pathogens by phagocytes (endocytosis), then the release of the hydrolyzed pieces of the pathogen outside the cell by exocytosis.

BULK TRANSPORT

- Endocytosis occurs when the plasma membrane envelops food particles and liquids.
- 1. **phagocytosis** – the cell takes in particulate matter
- 2. **pinocytosis** – the cell takes in only fluid
- 3. **receptor-mediated endocytosis** – specific molecules are taken in after they bind to a receptor

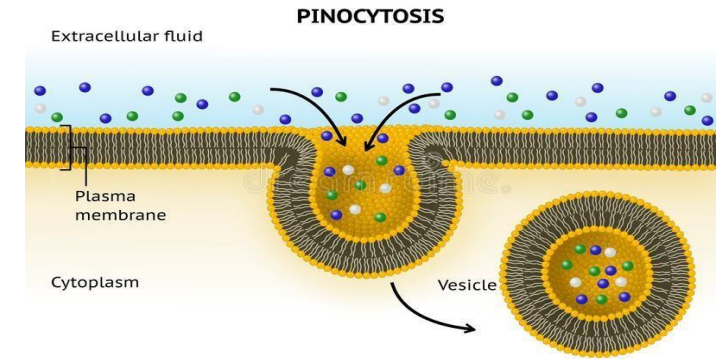
ENDOCYTOSIS

Endocytosis is defined as a transport mechanism by which the macromolecules enter the cell forming the vesicle into the cell. Macromolecules (substances with larger molecules) cannot pass through the cell membrane either by active or by passive transport mechanism. Such substances are transported into the cell by endocytosis.

Endocytosis is of three types:

1. Pinocytosis
2. Phagocytosis
3. Receptor-mediated endocytosis

Pinocytosis is



- Pinocytosis means ingestion of minute particles that form vesicles of extracellular fluid and particulate constituents inside the cell cytoplasm.
- In the process of pinocytosis the cell membrane forms an invagination.
- Whatever substance (Proteins) is found within the area of invagination is brought into the cell.
- In general this material will be dissolved in water and thus this process is also referred to as "cellular drinking".
- Pinocytosis is the only means by which most large macromolecules, such as most protein molecules, can enter cells. In fact, the rate at which pinocytotic vesicles form is usually

PINOCYTOSIS

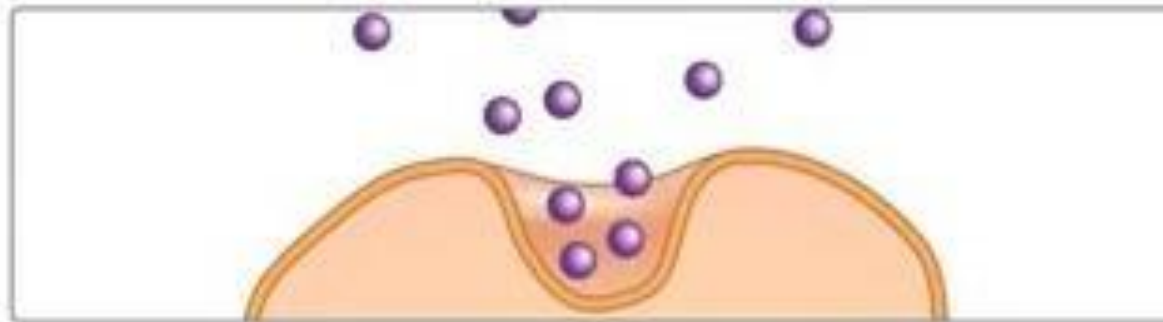
Pinocytosis is a process by which macromolecules like bacteria and antigens are taken into the cells. It is otherwise called the cell drinking.

Mechanism of pinocytosis

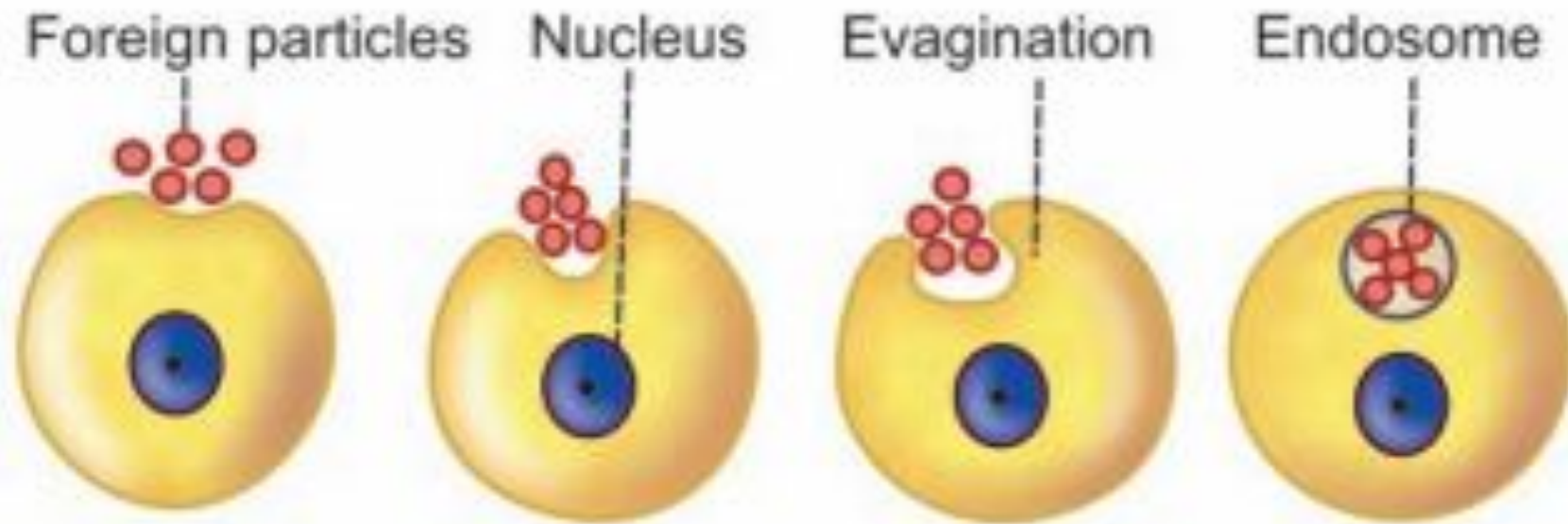
Pinocytosis involves following events:

- i). Macromolecules (in the form of droplets of fluid) bind to the outer surface of the cell membrane
- ii). Now, the cell membrane evaginates around the droplets
- iii). Droplets are engulfed by the membrane
- iv). Engulfed droplets are converted into vesicles and vacuoles, which are called endosomes
- v). Endosome travels into the interior of the cell
- vi). Primary lysosome in the cytoplasm fuses with endosome and forms secondary lysosome
- vii). Now, hydrolytic enzymes present in the secondary lysosome are activated resulting in digestion and degradation of the endosomal contents.

Pinocytosis



Process of Pinocytosis



PHAGOCYTOSIS

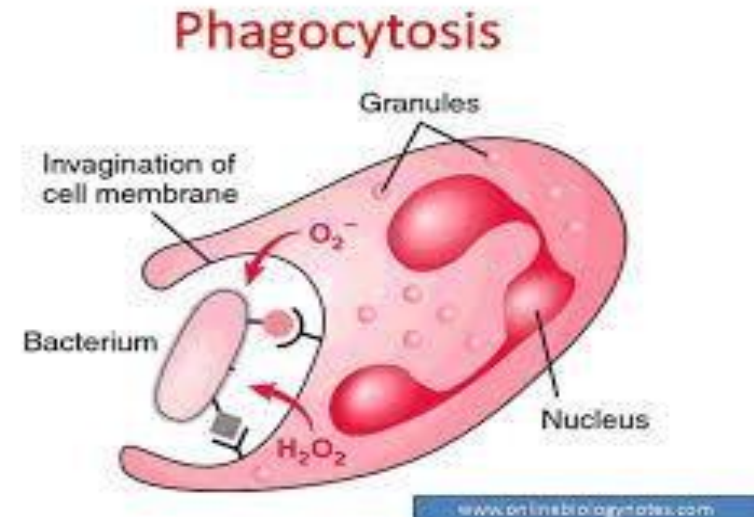
Phagocytosis is the process by which particles larger than the macromolecules are engulfed into the cells. It is also called cell eating. Larger bacteria, larger antigens and other larger foreign bodies are taken inside the cell by means of phagocytosis. Only few cells in the body like neutrophils, monocytes and the tissue macrophages show phagocytosis. Among these cells, the macrophages are the largest phagocytic cells.

Phagocytosis



Mechanism of Phagocytosis

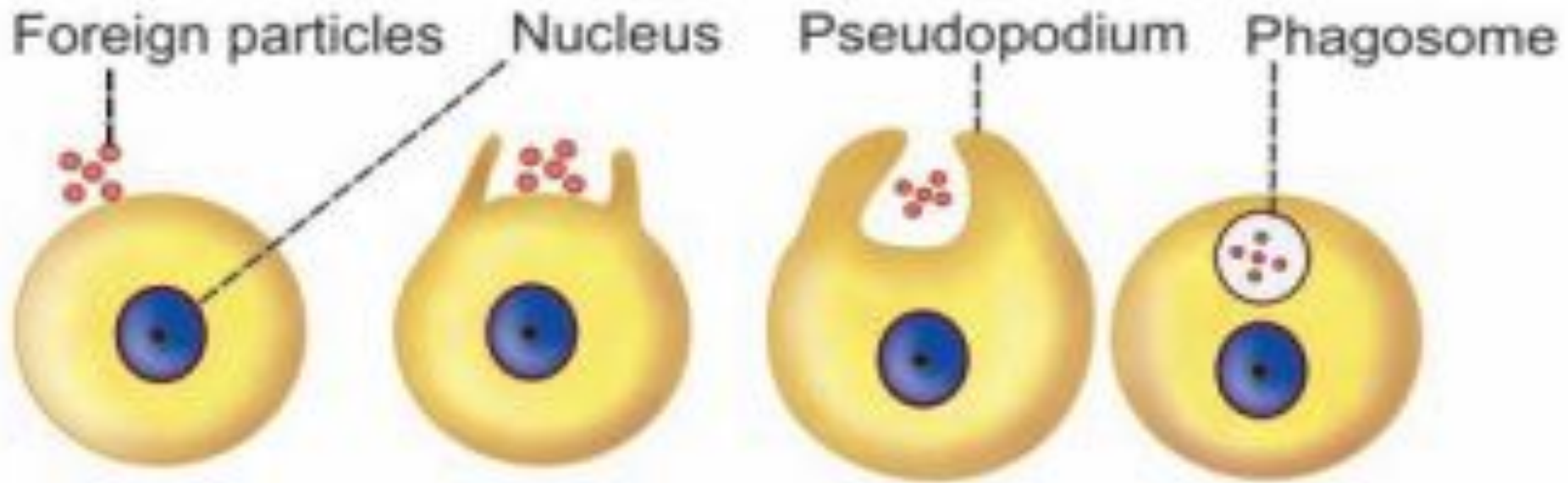
- The cell membrane invaginates the material from ECF.
- It is pinched off from the membrane and takes the material into ICF
- The phagocytic cell such as a macrophage may be attracted to a particle like a bacteria or virus by chemical attractant.



Steps of phagocytosis

- i. When bacteria or foreign body enters the body, first the phagocytic cell sends cytoplasmic extension (pseudopodium) around bacteria or foreign body
- ii. Then, these particles are engulfed and are converted into endosome like vacuole. Vacuole is very large and it is usually called the phagosome
- iii. Phagosome travels into the interior of cell
- iv. Primary lysosome fuses with this phagosome and forms secondary lysosome
- v. Hydrolytic enzymes present in the secondary lysosome are activated resulting in digestion and degradation of the phagosomal contents.

Process of Phagocytosis

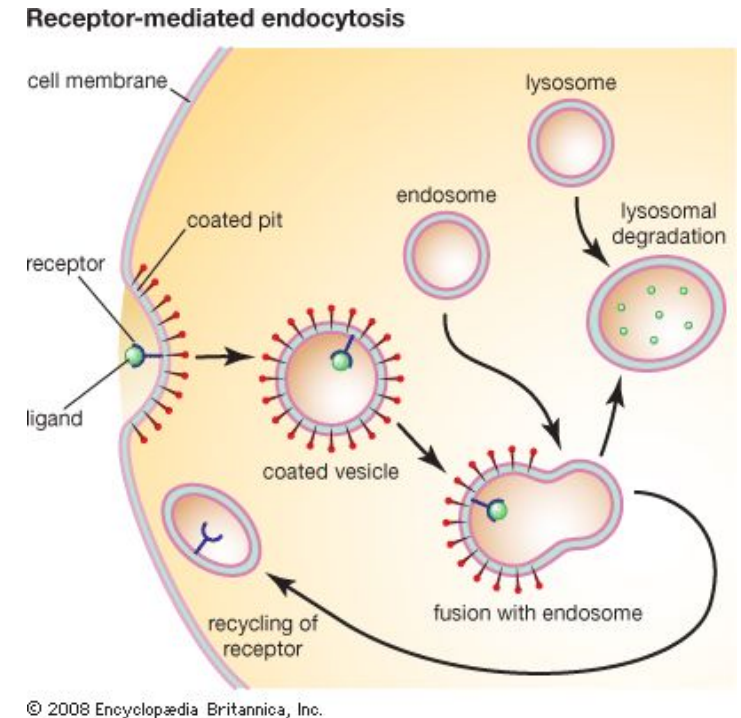


RECEPTOR MEDIATED ENDOCYTOSIS

Receptor-mediated endocytosis is the transport of macromolecules with the help of a receptor protein. Surface of cell membrane has some pits which contain a receptor protein called clathrin. Together with a receptor protein (clathrin), each pit is called receptor-coated pit. These receptor-coated pits are involved in the receptor mediated endocytosis.

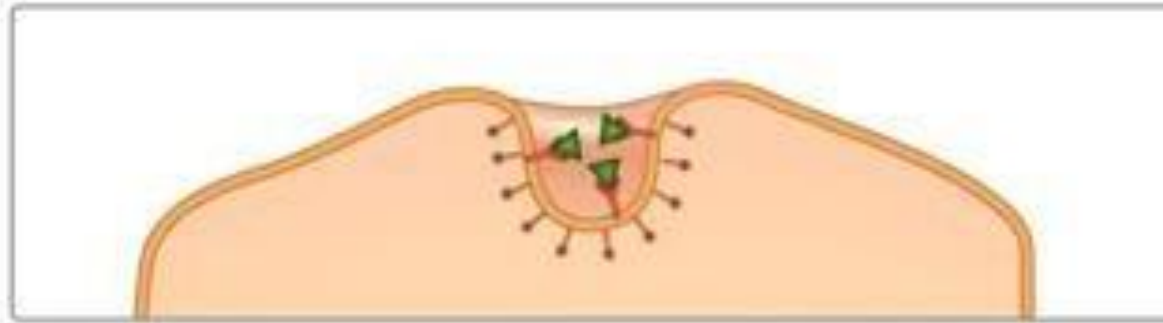
Receptor Mediated Endocytosis

- A selective process
- Involves formation of vesicles at surface of membrane
- Vesicles contain receptors on their membrane
- Vesicles contain specific target molecule in high concentration
- Clathrin-coated vesicle in cytoplasm
- uptake of LDL from bloodstream
- If receptors are lacking, LDL's accumulate and hypercholesterolemia develops



Receptor Mediated Endocytosis

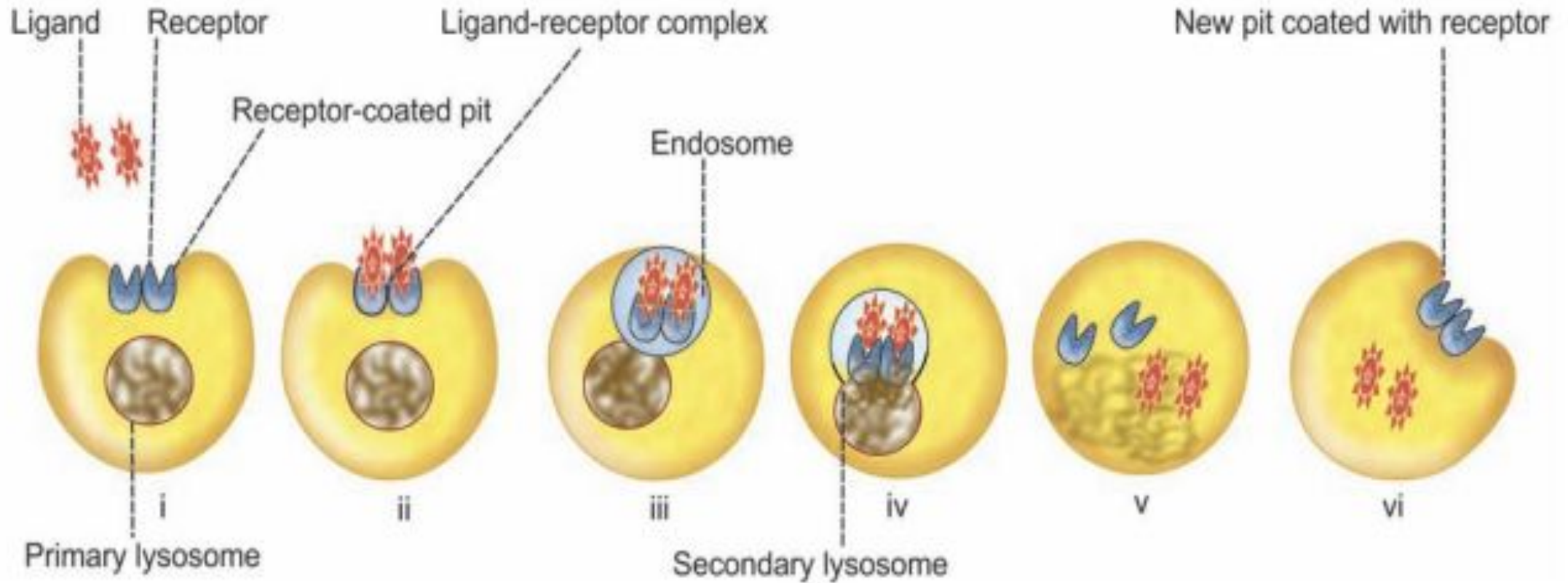
Receptor-mediated endocytosis



Mechanism of receptor-mediated endocytosis

- i. Receptor-mediated endocytosis is induced by substances like ligands
- ii. Ligand molecules approach the cell and bind to receptors in the coated pits and form ligand receptor complex
- iii. Ligand-receptor complex gets aggregated in the coated pits. Then, the pit is detached from cell membrane and becomes the coated vesicle. This coated vesicle forms the endosome
- iv. Endosome travels into the interior of the cell. Primary lysosome in the cytoplasm fuses with endosome and forms secondary lysosome.
- v. Now, the hydrolytic enzymes present in secondary lysosome are activated resulting in release of ligands into the cytoplasm
- vi. Receptor may move to a new pit of the cell membrane.

Mechanism of receptor mediated



Significance of Receptor mediated Endocytosis

Receptor-mediated endocytosis play an important role in the transport of several types of macromolecules into the cells, viz.

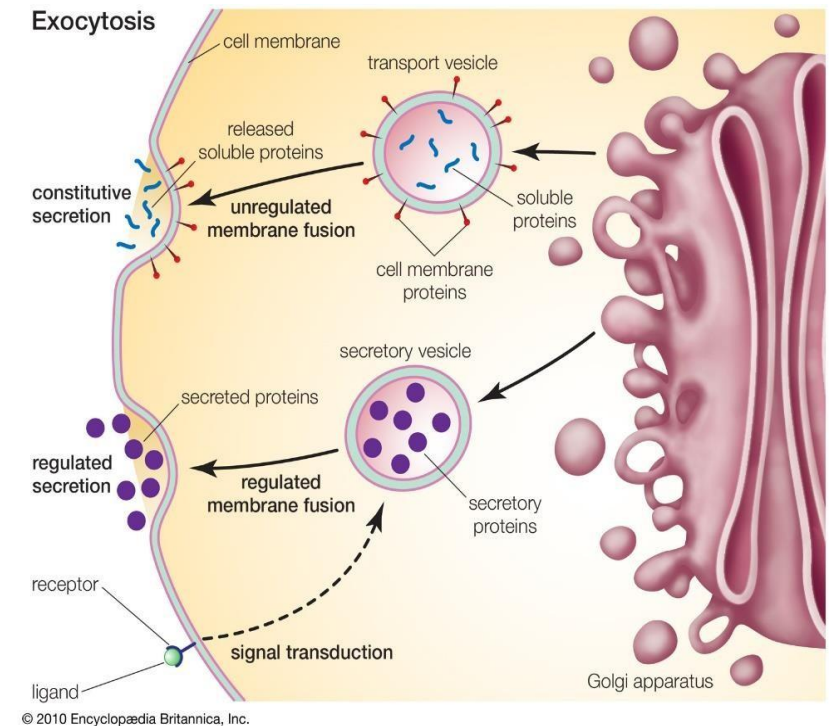
- i. Hormones:** Growth hormone, thyroid stimulating hormone, luteinizing hormone, prolactin, insulin, glucagon, calcitonin and catecholamines
- ii. Lipids:** Cholesterol and low-density lipoproteins (LDL)
- iii. Growth factors (GF):** Nerve GF, epidermal GF, platelet-derived GF, interferon
- iv. Toxins and bacteria:** Cholera toxin, diphtheria toxin, pseudomonas toxin, ricin and concanavalin A
- v. Viruses:** Rous sarcoma virus, semliki forest virus, vesicular stomatitis virus and adenovirus
- vi. Transport proteins:** Transferrin and transcobalamine
- vii. Antibodies:** IgE, polymeric IgG and maternal IgG.

Some of the receptor-coated pits in cell membrane are coated with another protein called caveolin instead of clathrin. Caveolin-coated pits are concerned with the transport of vitamins into the cell.

Exocytosis

is

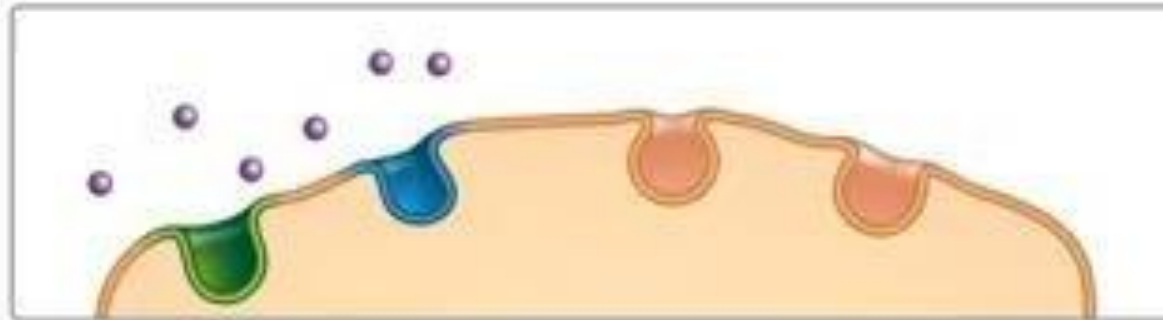
- Exocytosis is a process in which an intracellular vesicle (membrane bounded sphere) moves to the plasma membrane and fused the substance into the Extracellular fluids



Few of the processes that use Exocytosis are:

- 1. Secretion of proteins like enzymes and antibodies from cells.
- 2. Release of neurotransmitter from presynaptic neurons
- 3. Acrosome reaction during fertilization
- 4. Recycling of plasma membrane

Exocytosis



POST ASSESSMENT

1. Cell membrane is best defined as:
 - a) Completely permeable
 - b) Impermeable
 - c) Selectively permeable
 - d) Firm and rigid

2. Active transport:
 - a) Moves substances with concentration gradient
 - b) Requires ATP
 - c) Occurs only in plant cells
 - d) Does not involve membrane proteins

CONT..

3. Which process moves larger particles into the cell?

- a) Exocytosis
- b) Diffusion
- c) Endocytosis
- d) Osmosis

4. Channel proteins are important because they:

- a) Use ATP directly
- b) Allow specific molecules to cross the cell membrane
- c) Destroy harmful substances
- d) Make lipids



THANK
YOU